CASE FILE

FocusLock

Screen-time limits that follow you across Android and Windows.

PROBLEM

Screen-time limits are easy to dodge: switch browsers or use another device and the allowance resets.

DECISIONS

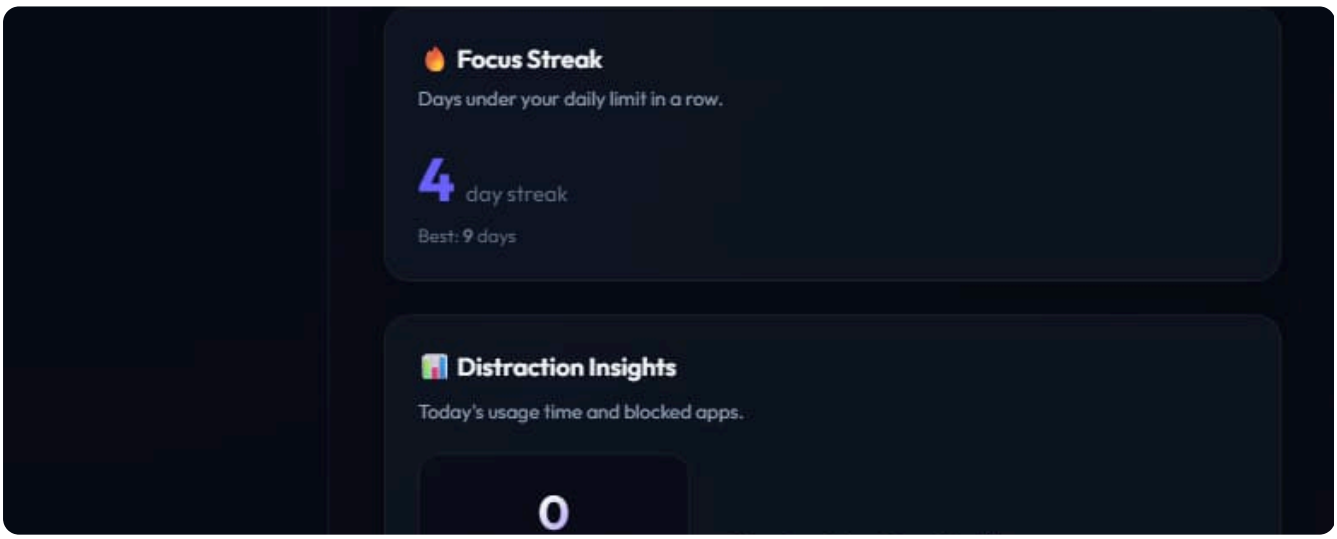
Enforce at the OS level, with one shared daily allowance. The first design was a self-hosted TypeScript backend plus a browser extension; I deleted it for Firestore as shared state and a native Electron agent, which removed a server and its firewall problem. Firestore web keys are public, so I wrote security rules that restrict each sync code to its members, capped at 10.

RESULT

One allowance across an Android phone (Accessibility service) and a Windows PC. A Ctrl+W bug, where the agent logged a closed tab that stayed open, was fixed by switching from PostMessage to SendInput. Over 250 automated tests; CI builds a signed APK plus Windows installer.

WHAT BROKE

The Windows agent logged that it had closed a tab, and the tab stayed open. PostMessage of WM_KEYDOWN for Ctrl never updates global key state, so Chromium read the synthetic W as a plain keypress with Ctrl not held. Fixed by re-asserting the foreground window and switching to SendInput, which injects input the OS actually accounts for.



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